Release Forecast

Velocity math, baseline KPI calculations, burndown forecast chart and 9-month window verdict

1. Given Project Data & Baseline Metrics

All calculations strictly use the project figures specified in the problem statement without inventing arbitrary numbers:

- **Total Product Backlog: W_total = 96 story points**.
- **Observed Team Velocity (First 3 Sprints, 2 Weeks Each): v_1 = 11 SP**, **v_2 = 14 SP**, **v_3 = 13 SP**.
- **Historical Venue Conflicts: 17 conflicts** across **320 events/year** (including two same-evening auditorium double-bookings).
- **Approval Turnaround Time:** Legacy average = **11 days**; Target = **3 days**.
- **Schedule & Budget Constraint: 9-month** developer graduation window (36–39 weeks); **Rs. 2.4 lakh** for 3 years of hosting/tools; **Rs. 0** for volunteer student developers.

2. Step-by-Step Velocity & Sprint Forecast Math

2.1 Cumulative Completed Work & Average Velocity

- **Work Completed in Sprints 1–3 (6 Weeks Elapsed): W_done = 11 + 14 + 13 = 38 story points**.
- **Average Velocity (v_avg): v_avg = (11 + 14 + 13) / 3 = 38 / 3 = 12.6667 ≈ 12.67 story points / sprint**.
- **Sample Standard Deviation (s): s = sqrt(((11 - 12.67)² + (14 - 12.67)² + (13 - 12.67)²) / (3 - 1)) = sqrt(4.6667 / 2) ≈ 1.53 SP / sprint**.

2.2 Remaining Work & Sprints Needed

- **Remaining Backlog Work (W_rem): W_rem = 96 - 38 = 58 story points**.
- **Exact Sprints Needed for Remaining Work (S_rem): S_rem = 58 / (38 / 3) = 174 / 38 ≈ 4.5789 sprints**.
- **Discrete Integer Sprints Needed: ceil(4.5789) = 5 additional sprints (Sprints 4, 5, 6, 7, and 8)**,equivalent to **10 additional weeks**.
- **Total Sprints to Finish All 96 Story Points: S_total = 3 + 4.58 = 7.58 sprints (15.16 weeks)** continuous, or **8 integer sprints = 16.0 weeks (~3.69 months)**.

3. Baseline Operational KPI Math (Approval Reduction & Venue Clash Rate)

Metric	Formula & Intermediate Steps	Exact Result	Operational Interpretation
Percentage Reduction in Approval Time Required	((T_current - T_target) / T_current) × 100 = ((11 - 3) / 11) × 100 = (8 / 11) × 100	72.73% reduction (8.0 days saved/event)	Saves 2,560 club-days of waiting per year across 320 events
Number of Venue Conflicts per Event	Conflicts_last_year / Events_per_year = 17 / 320	0.053125 conflicts/event (5.3125%)	1 venue clash every 18.82 events (Target in ClubSphere = 0.00)

4. Release Forecast Chart & Sprint Burndown Table

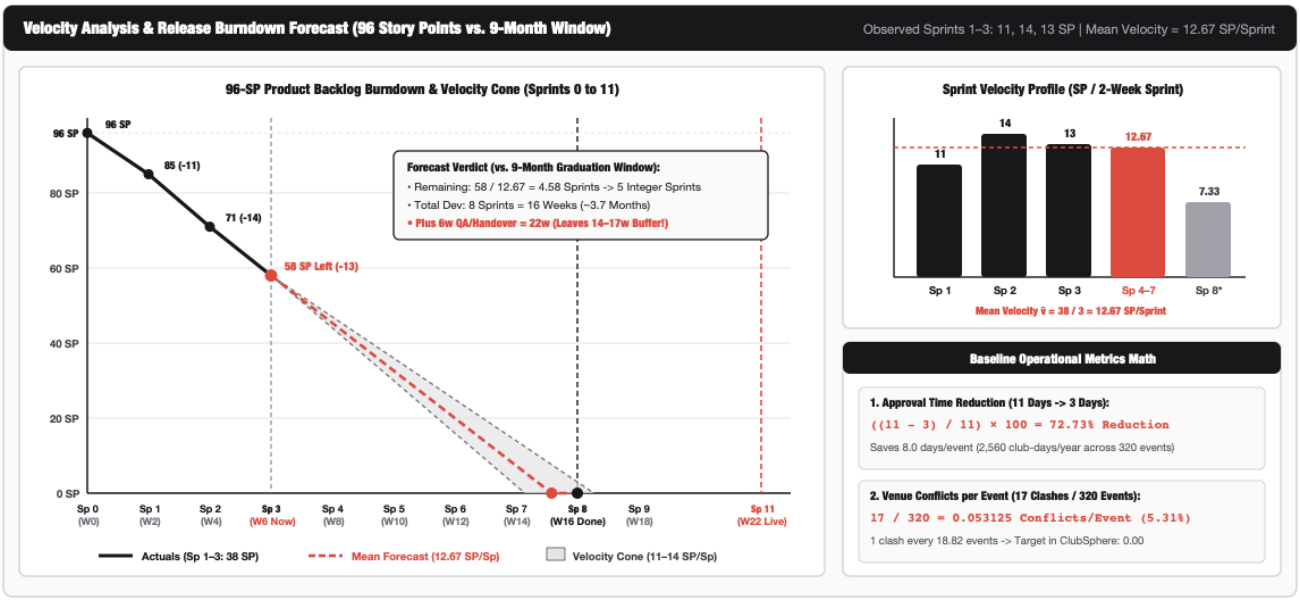


Figure 1 - Release burndown forecast cone (left), sprint velocity profile (top-right), and baseline KPI math (bottom-right).

Sprint	Timeline	Sprint Velocity	Cumulative Completed	Mean Remaining (v_avg=12.67)	Optimistic Rem (v=14)	Pessimistic Rem (v=11)
Sprint 0 (Start)	Week 0	—	0.00 SP	96.00 SP	96.00 SP	96.00 SP
Sprint 1 (Actual)	Week 2	11.00 SP	11.00 SP	85.00 SP	85.00 SP	85.00 SP
Sprint 2 (Actual)	Week 4	14.00 SP	25.00 SP	71.00 SP	71.00 SP	71.00 SP
Sprint 3 (Actual)	Week 6 (Now)	13.00 SP	38.00 SP	58.00 SP	58.00 SP	58.00 SP
Sprint 4 (Forecast)	Week 8	12.67 SP	50.67 SP	45.33 SP	44.00 SP	47.00 SP
Sprint 5 (Forecast)	Week 10	12.67 SP	63.33 SP	32.67 SP	30.00 SP	36.00 SP
Sprint 6 (Forecast)	Week 12	12.67 SP	76.00 SP	20.00 SP	16.00 SP	25.00 SP
Sprint 7 (Forecast)	Week 14	12.67 SP	88.67 SP	7.33 SP	2.00 SP	14.00 SP
Sprint 8 (Forecast)	Week 16 (3.7m)	7.33 SP (cap 12.67)	96.00 SP (100%)	0.00 SP (Done at Sp 7.58)	0.00 SP (Sp 7.14)	3.00 SP
Sprint 9 (Worst Case)	Week 18 (4.2m)	—	96.00 SP	0.00 SP	0.00 SP	0.00 SP (Done at Sp 8.27)

5. Verdict on the 9-Month Window & Why an Estimate Is Not a Promise

5.1 Verdict on the 9-Month (36–39 Week) Graduation Window

Verdict: HIGHLY FEASIBLE WITH A 14-TO-17-WEEK SAFETY MARGIN. Completing all 96 story points requires **8 sprints (16 weeks / 3.69 months)** at the mean velocity of 12.67 SP/sprint (or 9 sprints / 18 weeks under the worst observed velocity of 11 SP/sprint). Adding 3 weeks for System QA & 46-club UAT pilot (Weeks 17–19) and 3 weeks for production go-live & junior maintainer handover (Weeks 20–22) brings total project duration to **22 weeks (~5.08 months)**, leaving **14–17 weeks of buffer** before graduation.

5.2 Key Assumptions & Why This Forecast Is an Estimate, Not a Promise

- **1. Small Sample Variance (n = 3 Sprints):** Deriving a mean from 3 sprints (11, 14, 13) yields a 95% confidence interval of 12.67 ± 3.80 SP/sprint. Early setup stories differ from complex concurrency and PDF generation work in later sprints.
- **2. Non-Stationary Student Availability:** Volunteer developers receive Rs. 0 pay and face university mid-term exams, semester finals, and campus placement drives that temporarily dip sprint capacity.
- **3. Emergent Complexity & Defect Rework:** Velocity measures gross story points completed; if integration testing uncovers venue race conditions or CSV encoding bugs, capacity must shift to bug remediation.
- **4. Goodhart's Law:** Treating 12.67 SP/sprint as a rigid promise incentivizes point inflation or skipping handover documentation. Instead, we manage uncertainty via MoSCoW prioritization and a 14-week pre-graduation buffer.